# ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** cork and leaf | Cork and leaf are lighter than water. |
| 2 | **(d)** the boat's shape holds a lot of air | Large volume with air makes the boat's overall density small. |
| 3 | **(c)** food does not stick to the surface | Non-stick. |
| 4 | **(b)** salt and sugar | Both dissolve. |
| 5 | **(b)** wax melts quickly along the rod when one end is heated | Heat travels along it. |
| 6 | **(b)** wool is a poor conductor of heat | Traps body heat. |
| 7 | **(b)** metal | Metals have these properties. |
| 8 | **(d)** a thick wooden handle | Wood does not conduct heat. |
| 9 | **(b)** steel | Strong and waterproof. |
| 10 | **(a)** thin cotton cloth | Translucent cloth. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | The boat has a large hollow shape with air (1); Water supports it as its average density is low (1). | 1+1 |
| 12 | No privacy (1); Frosted translucent glass is better (1). | 1+1 |
| 13 | Both disappear on stirring (1); Both are soluble (1). | 1+1 |
| 14 | Steel is stronger and harder than pure iron (1); Stainless steel does not rust easily (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Magnet removes iron (1); Dissolve sugar in water (1); Filter sand and evaporate water for sugar (1). | 1+1+1 |
| 16 | Rough mat cleans shoes (1); Smooth top is easy to clean (1); Window lets in light (1). | 1+1+1 |
| 17 | Thermocol is light (1); Combined object has low overall density (1); It floats (1). | 1+1+1 |
| | **OR:** Heat one end of each (1); Place wax pins at equal distance (1); The pin that falls first shows the best conductor (1). | 1+1+1 |
| 18 | Cut a piece of cork (1); Even small solid pieces float (1); Hence lighter material (1). | 1+1+1 |
| | **OR:** Pins conduct (1); Plastic insulates (1); Safe for users (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Heat water and cover (1); Measure temperature drop after 30 min (1); Same conditions for all (1); The least drop is best insulator (1); Cork or plastic likely (1). | 1+1+1+1+1 |
| | **OR:** Dissolve salt in water (1); Fold filter paper (1); Pour mixture (1); Sand stays on paper (1); Salt solution passes (1). | 1+1+1+1+1 |
| 20 | Heat a rod for conduction (1); Stir in water for solubility (1); Torch for transparency (1); Put in water for floating (1); Scratch for hardness (1). | 1+1+1+1+1 |
| | **OR:** Animal: wool, silk (1); Plant: cotton, wood (1); Mineral: iron, stone (1); Man-made: plastic (1); Natural sources give raw materials (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Calling frosted glass or tracing paper transparent; they are **translucent**.
- Saying that insoluble means the substance cannot mix with anything; it means it does not dissolve in **water**.
- Writing that all metals are hard; sodium is soft, and only most metals are hard.
- Confusing **opaque** with translucent: opaque blocks all light.
- Thinking oil is soluble because it spreads on water; it floats as a separate layer.

- Assuming all shiny objects are metals; lustre alone is not proof, check conduction too.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** wooden chips | Wood is non-magnetic. |
| 2 | **(c)** can be made into thin sheets | Metals are malleable. |
| 3 | **(a)** wood | Wood burns. |
| 4 | **(c)** cooking oil | Oils do not mix. |
| 5 | **(d)** breaks easily on hitting | Glass. |
| 6 | **(d)** glass | Glass breaks when hit. |
| 7 | **(a)** they form shadows | They block light. |
| 8 | **(c)** elastic and soft | Rubber rubs off marks. |
| 9 | **(b)** plastic | Plastic. |
| 10 | **(a)** spectacles | Clear view. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Heat is conducted along the rod (1); It reaches nearer points first (1). | 1+1 |
| 12 | Each part needs a different property (1); A pan has a metal body and plastic handle (1). | 1+1 |
| 13 | Wool traps air (1); Air is a poor conductor of heat (1). | 1+1 |
| 14 | It conducts electricity well (1); It is flexible and easy to draw (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Ceramic is a poor conductor (1); Does not burn fingers (1); Keeps tea hot (1). | 1+1+1 |
| 16 | Blocks light (1); Keeps heat out (1); Gives privacy (1). | 1+1+1 |
| 17 | Cloth for body (1); Plastic zip (1); Metal buckle (1). | 1+1+1 |
| | **OR:** Glass: transparent, breakable (1); Plastic: light, does not break (1); Plastic is cheaper (1). | 1+1+1 |
| 18 | Raincoat needs a waterproof material (1); Cooking pan needs a heat-conducting metal (1); Properties decide use (1). | 1+1+1 |
| | **OR:** Polyester absorbs less water (1); Water stays on the surface (1); Evaporates quickly (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Pan: conducts heat (1); Tyre: elastic (1); Window: transparent (1); Bridge: strong (1); Raincoat: waterproof (1). | 1+1+1+1+1 |
| | **OR:** It breaks easily (1); It is heavy (1); Plastic is lighter (1); Metals are stronger (1); Choose by use (1). | 1+1+1+1+1 |
| 20 | Copper: wires (1); Aluminium: foil (1); Glass: windows (1); Rubber: tyres (1); Wood: furniture (1). | 1+1+1+1+1 |
| | **OR:** Conductors: copper (1); Uses: vessels, rods (1); Insulators: wood, plastic (1); Uses: handles, flasks (1); Choose based on need (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Calling frosted glass or tracing paper transparent; they are **translucent**.
- Saying that insoluble means the substance cannot mix with anything; it means it does not dissolve in **water**.
- Writing that all metals are hard; sodium is soft, and only most metals are hard.
- Confusing **opaque** with translucent: opaque blocks all light.
- Thinking oil is soluble because it spreads on water; it floats as a separate layer.
- Assuming all shiny objects are metals; lustre alone is not proof, check conduction too.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** flexible | Paper, wood. |
| 2 | **(c)** rubber | Rubber does not conduct electricity. |
| 3 | **(d)** bathroom windows | Provides privacy with some light. |
| 4 | **(c)** it gets soggy and weak | Paper absorbs water. |
| 5 | **(d)** strong and hard | Strong. |
| 6 | **(d)** it is lustrous and does not rust | Shiny and unreactive. |
| 7 | **(a)** a rubber tyre | Rubber is elastic and strong. |
| 8 | **(b)** absorbs sweat | Absorbs water. |
| 9 | **(c)** it is transparent but takes up space | Air is transparent. |
| 10 | **(b)** jute, nylon | Fibres are flexible. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Rubber gives good grip (1); It is flexible and durable (1). | 1+1 |
| 12 | Container: steel or plastic (1); Cloth: cotton; tyre: rubber (1). | 1+1 |
| 13 | Cotton is from plants (1); Nylon is made in factories (1). | 1+1 |
| 14 | Iron conducts heat quickly (1); Wood is a poor conductor (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Wood: easy to shape (1); Graphite: marks paper (1); Rubber: erases; metal ferrule holds it (1). | 1+1+1 |
| 16 | Wood is lighter than the same volume of water (1); Steel is heavier than the same volume of water (1); So the wooden ruler floats and the steel ruler sinks (1). | 1+1+1 |
| 17 | Natural: leather, wool, cotton (1); Man-made: plastic, nylon (1); Polythene is also man-made (1). | 1+1+1 |
| | **OR:** Material is the substance (1); Object has a shape and use (1); Same material can make different objects (1). | 1+1+1 |
| 18 | Add water and stir (1); Sugar dissolves (1); Chalk remains as sediment (1). | 1+1+1 |
| | **OR:** Pour water on it (1); Observe if it passes (1); Check if it soaks (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Metal base conducts heat (1); Plastic handle insulates (1); Each part has a job (1); Combined object is safe (1); Both properties are used (1). | 1+1+1+1+1 |
| | **OR:** Sorting by material (1); Metals recycled (1); Glass and paper recycled (1); Saves resources (1); Reduces waste (1). | 1+1+1+1+1 |
| 20 | Cork, wood, bottle float (1); Coin, stone, nail sink (1); Floating objects are lighter than water (1); Sinking are heavier (1); Shape can matter with hollow things (1). | 1+1+1+1+1 |
| | **OR:** Does not decay (1); Pollutes land and water (1); Jute bags (1); Paper and cloth bags (1); Reuse plastic (1). | 1+1+1+1+1 |

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